SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1710**

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4 **Total Marks:40**

Annexure A

CASE STUDY

Code of Conduct:

I V. RUPASHREE (192319012) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: V. Rup.

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Student 28: RUPASHREE V | Register Number: 192319012

Question 1

Case Study - Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

I. If a patient has Fever AND Cough -> Possible Flu
Ii. If a patient has Fever AND Rash -> Possible Measles
Iii. If a patient has Cough AND Breathlessness -> Possible Pneumonia
Iv. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

(a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.

(b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.

(c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Question 2

Case Study - Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

I. for-all x: Vehicle(x) AND EmergencyType(x) -> ClearPath(x) [Rule 1]
II. for-all x: ClearPath(x) -> GreenSignal(x) AND NOT RedSignal(x) [Rule 2]
III. for-all x for-all y: Vehicle(x) AND Behind(x,y) AND GreenSignal(y) -> Proceed(x) [Rule 3]
IV. Facts: Vehicle(Ambulance), EmergencyType(Ambulance), Vehicle(CarA), Behind(CarA, Ambulance)

(a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set theta (MGU) at each step.

(b) Using the Resolution method, prove that 'Proceed(CarA)' is true. Convert relevant clauses to CNF and show each resolution step.

(c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

Question 3

Case Study - Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

1) P1: SoilDry -> IrrigationNeeded
2) P2: IrrigationNeeded AND CropWheat -> ApplyDripMethod
3) P3: NOT ApplyDripMethod AND CropWheat -> CropAtRisk
4) Facts: SoilDry = True, CropWheat = True

(a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication -> move NOT inward -> distribute).

(b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause. Draw the resolution proof tree.

(c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

Question 4

Case Study - Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

1) Stench[1,2] -> Wumpus is adjacent to cell [1,2]
2) Breeze[1,1] -> Pit is adjacent to cell [1,1]
3) Glitter[x,y] -> Gold is at [x,y]
4) NOT Wumpus[x,y] AND NOT Pit[x,y] -> Safe[x,y]

(a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

(b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

(c) Evaluate whether the path [1,1] -> [1,2] -> [2,2] is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks (100)
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly	24
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	24
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	19
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	19
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	9

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CO3 - AT2

Smart Medical diagnosis System:-

(a) Proportional Logic Regression:-

Let:

F = Patient A has Fever

C = Patient A has cough

R = Patient A has Rash

B = Patient A has breathlessness

Flu = Possible Flu

Measles: Possible Measles

Pneumonia: Possible Pneumonia

Rules:-

$F \wedge C \rightarrow Flu$

$F \wedge R \rightarrow Measles$

$C \wedge B \rightarrow Pneumonia$

Facts:-

F = True

C = True

B = True

(b) Forward Chaining:

Initial facts:

F, C, B

Step 1:

$F \wedge C \rightarrow$ flue

F = True and C = True

Therefore,

Flu = True

Step 2:

$F \wedge R \rightarrow$ Measles

R is not given, so this rule cannot fire.

Step 3:

$C \wedge B \rightarrow$ Pneumonia

C = True and B = True

Therefore

Pneumonia = True

Final conclusion:-

Possible Flue = True

Possible Pneumonia = True

Possible Measles cannot be derived.

(c) Backward Chaining:-

Goal:

Pneumonia

$C \wedge B \rightarrow$ Pneumonia

Pneumonia

↙ ↘

C and B

↓ ↓

C B

↓ ↓

Fact Fact

(Since both are facts)

∴ Pneumonia is proved.

2) Autonomous Traffic Management Agent:-

(a) Unification:-

Rule 1:

$\forall x$ [Vehicle (x) ^ Emergency Type (x) → Clear path (x)]

Facts:-

Vehicle (Ambulance)

Emergency Type (Ambulance)

Match:

Vehicle (x) with Vehicle (Ambulance)

Therefore:

$\theta_1 = \{x / Ambulance\}$

Match:

Emergency Type (x) with Emergency Type (Ambulance)

Therefore,

$\theta_2 = \{x / Ambulance\}$

Applying Substitution:

ClearPath (Ambulance)

(b) Resolution proof of Proceed (Car A)

Relevant rules:

R_1:

Vehicle (x) ∧ EmergencyType (x) → ClearPath (x)

CNF:

¬ Vehicle (x) ∨ ¬ EmergencyType (x) ∨ ClearPath (x)

R_2

ClearPath (x) → GreenSignal (x)

CNF:

¬ ClearPath (x) ∨ GreenSignal (x)

R_3:

Vehicle (x) ∧ Behind (x, y) ∧ GreenSignal (y) → Proceed (x)

CNF:

¬ Vehicle (x) ∨ ¬ Behind (x, y) ∨ ¬ GreenSignal (y) ∨ Proceed (x)

Facts:

Vehicle (Ambulance)

EmergencyType (Ambulance)

Vehicle (Car A)

Behind (Car A, Ambulance)

Additional rules / facts are required, such as :

Priority (Emergency Vehicle, level)

and

HigherPriority(x, y) ^ EmergencyVehicle(x) ^
EmergencyVehicle(y) → Give Priority(x)

∴ The KB should be extended with priority & conflict resolution rules to handle simultaneously emergencies logically.

3) Agricultural Expert Reasoning System:-

(a) CNF:-

1. SoilDry → IrrigationNeeded → ¬ SoilDry ∨ Irrigation Needed
2. Irrigation Needed ∧ Crop ~~Wheat~~ → Apply Drip Method
 → ¬ IrrigationNeeded ∨ ¬ CropWheat ∨ Apply drip Method.
3. ¬ Apply Drip Method ∧ Crop Wheat → Crop at Risk.

(b) Resolution:-

SoilDry → IrrigationNeeded + Crop wheat → Apply drip method.

Negated goal : ¬ Apply Drip Method

∴ ApplyDripMethod is proved.

Resolution:

1. From R_1 & the two Ambulance facts:

ClearPath (Ambulance)

2. From R_2:

Clearpath (Ambulance) $\longrightarrow$ GreenSignal (Ambulance)

3. R3 with:

- Vehicle (car A)
- Behind (Car A, Ambulance)
- GreenSignal (Ambulance)

gives:

Proceed (Car A)

Hence,

$\therefore$ Proceed (Car A) is True.

(c) Two simultaneously emergency vehicles:-

No, the current KB is not completely sufficient.

The current rules can identify an emergency vehicle fast and clear its path, but there is no explicit rule for

- Prioritizing two emergency vehicles, or
- Resolving conflicts between their paths,
- assigning different lanes / routes,
- deciding which emerging vehicle gets priority.

If sailDry is removed:-

*IrrigationNeeded cannot be derived.

* ApplyDripMethod cannot be derived.

→ApplyDripMethod is also not given.

∴ CropAtRisk cannot be concluded.

4) Cave Exploration:-

(a) Belief State:-

Given:

Draft [1, 3]

Glow [1,1]

Sparkle [2,3]

Wind means:-

Pit adjacent to [1, 3]

Lavavent adjacent to [1, 1]

Treasure at [2, 3]

(b) Forward Chaining:-

Draft[1, 3] → Pit near [1, 3]

Glow [1, 1] → Lavavent near [1, 1]

Sparkle [2,3] → Treasure [2,3]

(c) Path Safety :-

Path: [1,1] → [1, 3] → [2,3]